Health360: Comprehensive Healthcare Risk & Disease Analysis

Final project report · Data Analytics with Tableau

Prepared by: Nihar Vijayvargiya

1. Introduction

1.1 Project Overview

Health360 turns 5,000 patient records into an interactive Tableau dashboard and data story that show which patient groups are most exposed to heart disease, diabetes and stroke. The dataset covers age, gender, BMI, exercise level, smoking, alcohol, blood pressure, cholesterol, glucose, symptoms and disease flags. The finished dashboard and story are published on Tableau Public and embedded in a Flask web app.

1.2 Purpose

The project helps clinicians, analysts and public health researchers spot high-risk segments early, see how lifestyle and demographic factors relate to disease, and plan preventive care.

2. Ideation Phase

2.1 Problem Statement

Chronic diseases such as heart disease, diabetes and stroke are often detected late, and reviewing patient records by hand hides patterns. Three users share this problem:

- Dr. Michael Carter, Clinical Risk Assessment Officer: needs to find high-risk patients early from health indicators and KPIs.
- Emily Rodriguez, Healthcare Data Analyst: needs to see how smoking, alcohol and exercise relate to chronic disease.
- Dr. Priya Nair, Public Health Researcher: needs to study age-group, BMI and symptom patterns across the population.

2.2 Empathy Map Canvas

Persona: Dr. Michael Carter, Clinical Risk Assessment Officer.

Quadrant	Notes
Says	"I need to see high-risk patients before complications start."
Thinks	Late detection raises treatment cost and harms outcomes.
Does	Reads patient records and indicators such as BMI, blood pressure, cholesterol and

Quadrant	Notes
	glucose.
Feels	Frustrated by slow manual review; wants quick, trustworthy views.
Pain	Patterns across thousands of records are hard to see by hand.
Gain	A dashboard with KPIs and filters that points to at-risk groups.

2.3 Brainstorming

- Show headline KPIs (patients, diabetes, heart disease, stroke) at the top.
- Group raw ages into six age groups so segments are easy to compare.
- Group BMI into four categories to link body weight with disease rates.
- Use a heat map to find combinations of BMI and exercise level with the highest heart disease rate.
- Add filters for BMI Category, Age Group and Exercise Level.
- Tell the findings as a story, then embed dashboard and story in a web page.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	What the user does	Need
Open	Opens the Health360 web page	Clear overview and headline numbers
Explore	Uses Age Group, BMI Category and Exercise Level filters	Fast, interactive filtering
Compare	Reads charts of blood pressure, lifestyle and disease rates	Easy-to-read visuals
Understand	Follows the story scene by scene	A clear narrative
Act	Picks groups for screening and prevention	Reliable insight

3.2 Solution Requirement

- Load all 5,000 records into Tableau without truncation.
- Clean and group data with calculated fields (Age Group, BMI Category).
- Build 9 distinct visualizations.
- Combine them into a responsive dashboard with interactive filters.
- Build a story with one scene per chart.
- Publish to Tableau Public and embed in a Flask web app.

3.3 Data Flow Diagram

1. Dataset (CSV on Google Drive) is downloaded.
2. Tableau Desktop connects to the CSV and prepares the data.
3. Visualizations, dashboard and story are built.
4. The workbook is published to Tableau Public.
5. The Flask app embeds the dashboard and story.
6. The user explores the insights in a browser.

3.4 Technology Stack

Layer	Tool
Data source	CSV file (Google Drive)
Analysis and visualization	Tableau Desktop
Hosting	Tableau Public
Web app	Flask (Python), HTML, CSS
Code hosting	GitHub

4. Project Design

4.1 Problem Solution Fit

Problem	Solution
Manual review hides disease patterns	Charts and a heat map show patterns at a glance
Raw age and BMI values are hard to compare	Calculated fields group them into clear categories
Insights stay inside the analyst's tool	A published, embedded dashboard reaches any browser

4.2 Proposed Solution

A Tableau dashboard and story built on the smart healthcare dataset, with nine visualizations, calculated fields for Age Group and BMI Category, and three filters. The workbook is published to Tableau Public and shown inside a Flask page with Home, About, Dashboard, Story and FAQ sections.

4.3 Solution Architecture

Google Drive (dataset) → Tableau Desktop (connection, preparation, charts, dashboard, story) → Tableau Public (publishing) → Flask web app (embedding) → End user.

5. Project Planning & Scheduling

5.1 Project Planning

Estimated time per SkillWallet epic: Data Collection 5 h, Data Preparation 5 h, Data Visualization 5 h, Project Demonstration & Documentation 5 h.

Epic	Activities
Data Collection & Extraction	Download the dataset; connect it to Tableau
Data Preparation	Check missing values and duplicates; create Age Group and BMI Category fields
Data Visualization	Build 9 unique visualizations
Dashboard	Combine charts with filters in a responsive layout
Story	One scene per chart
Performance Testing	Data loaded, filters, calculation fields, number of visualizations
Web Integration	Publish to Tableau Public; embed in Flask
Demonstration & Documentation	Record 5–7 minute video; write this report

6. Functional and Performance Testing

6.1 Performance Testing

Check	Result
Amount of data rendered	5,000 rows loaded, confirmed by the Count of Patients KPI
Utilization of data filters	Age Group, BMI Category and Exercise Level filters
Number of calculation fields	3 (Age Group, Bmi Category, Exercise Label)
Number of visualizations	9

Data checks: no missing values and no duplicate rows. The health_risk_score column holds the value 100 in every row, so it was not used.

7. Results

7.1 Output Screenshots

Screenshots of each visualization, the dashboard, the story and the Flask web app.

Figure 1. Health360 dashboard in Tableau, full view with KPI cards: 5,000 patients, 1,746 diabetes, 1,499 heart disease, 1,243 stroke.

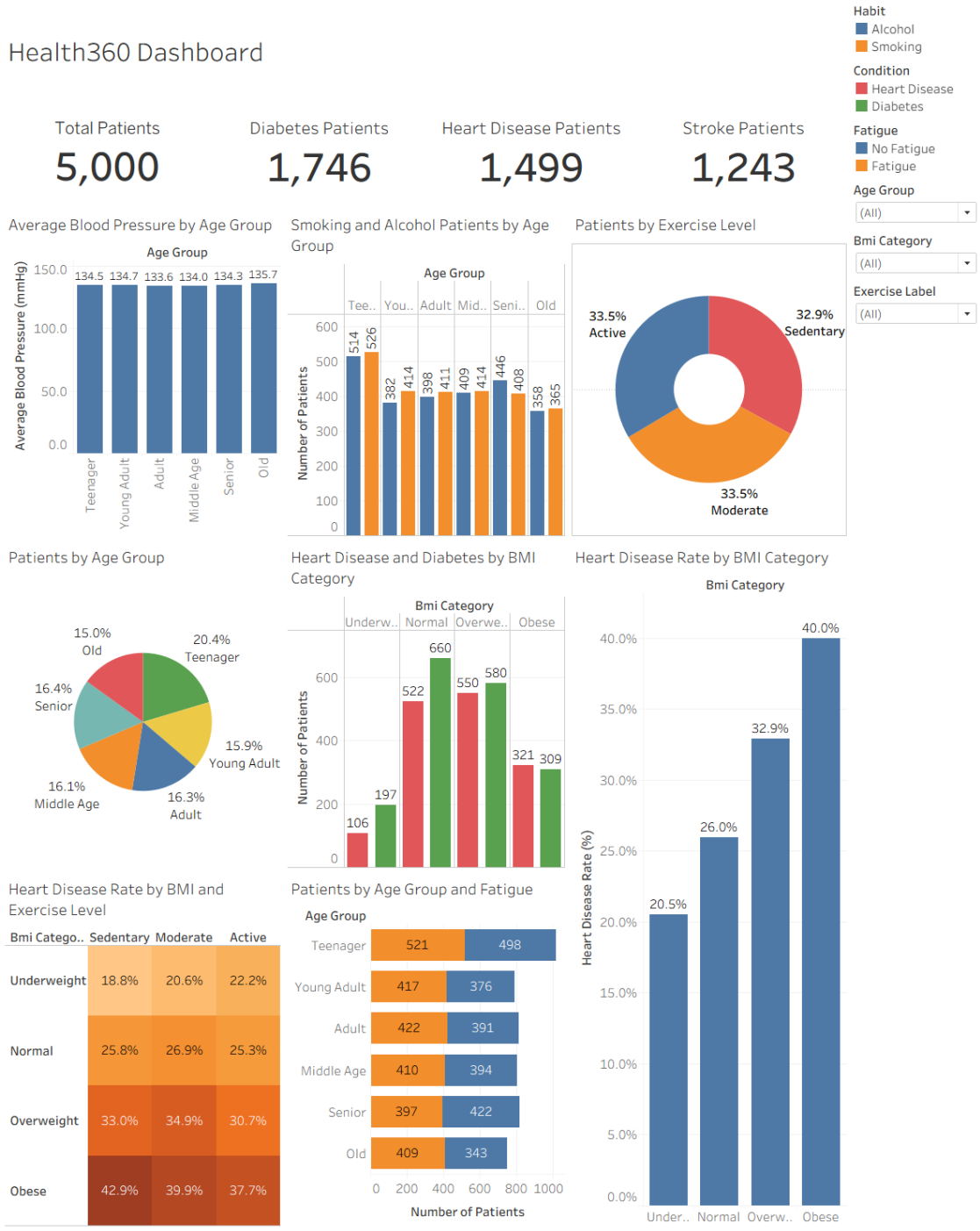


Figure 2. Average Blood Pressure by Age Group: 133.6 to 135.7 mmHg, a gap of about 2 mmHg.

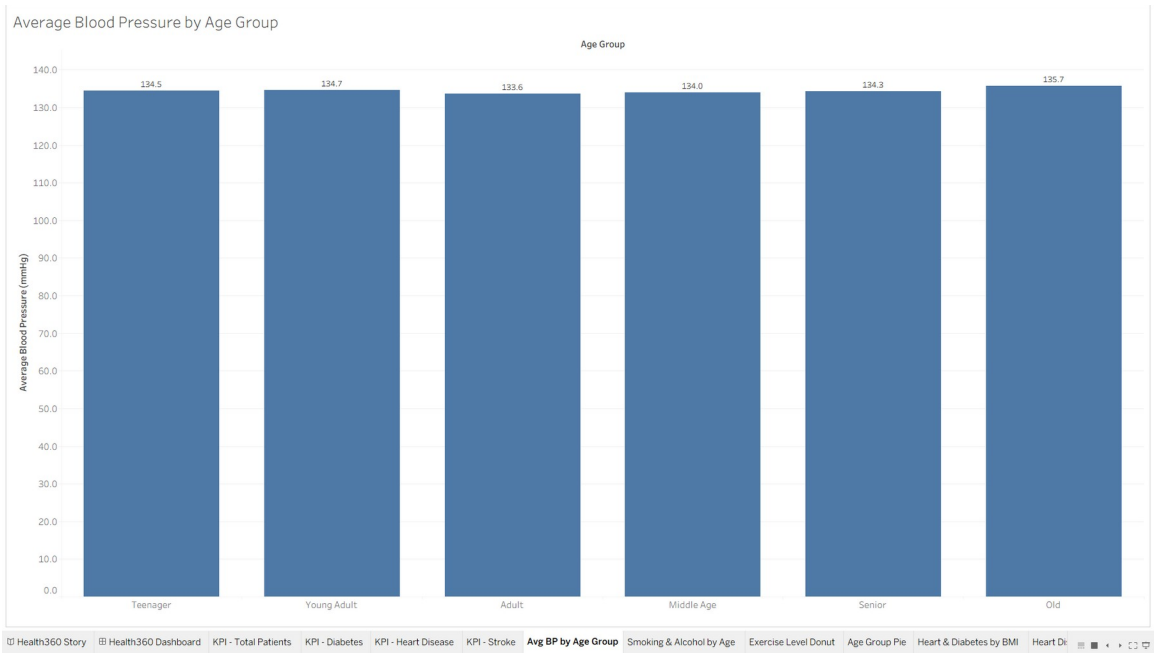


Figure 3. Smoking and Alcohol Patients by Age Group.

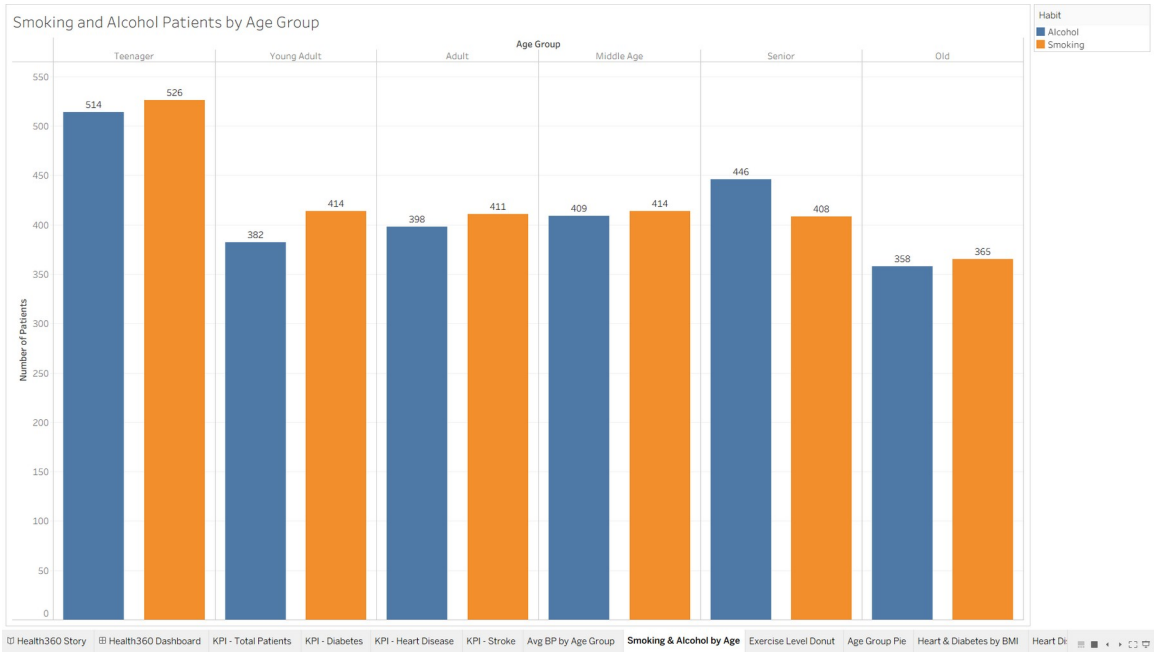


Figure 4. Patients by Exercise Level: Moderate 33.5%, Active 33.5%, Sedentary 32.9%.

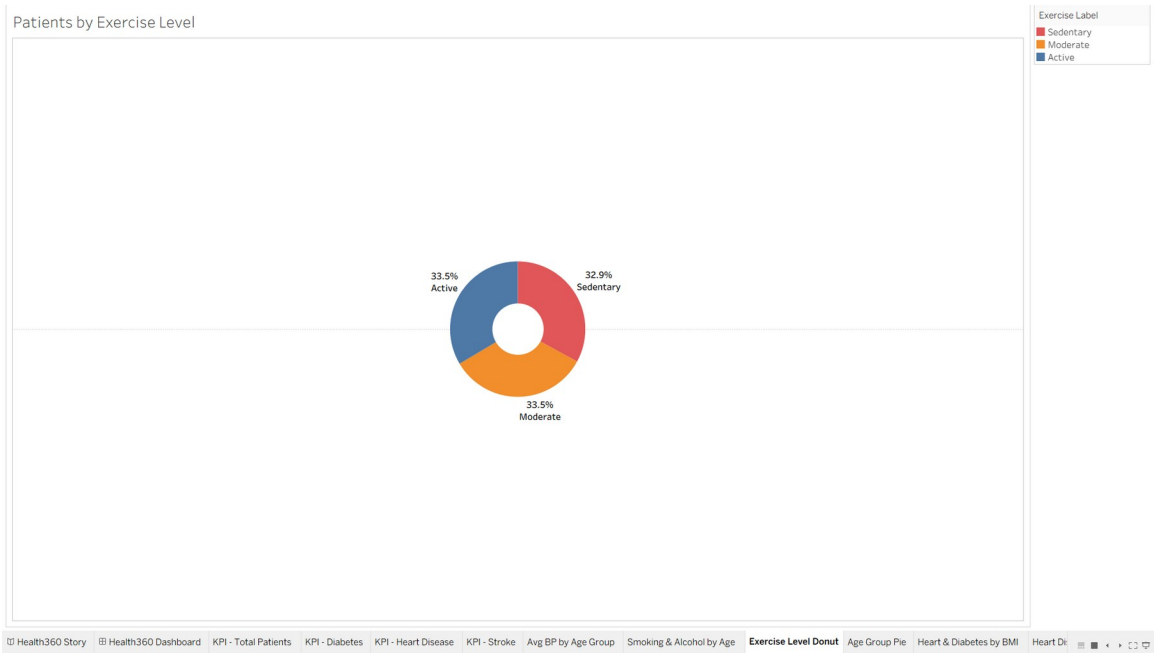


Figure 5. Patients by Age Group: Teenager 20.4%, Senior 16.4%, Adult 16.3%, Middle Age 16.1%, Young Adult 15.9%, Old 15.0%.

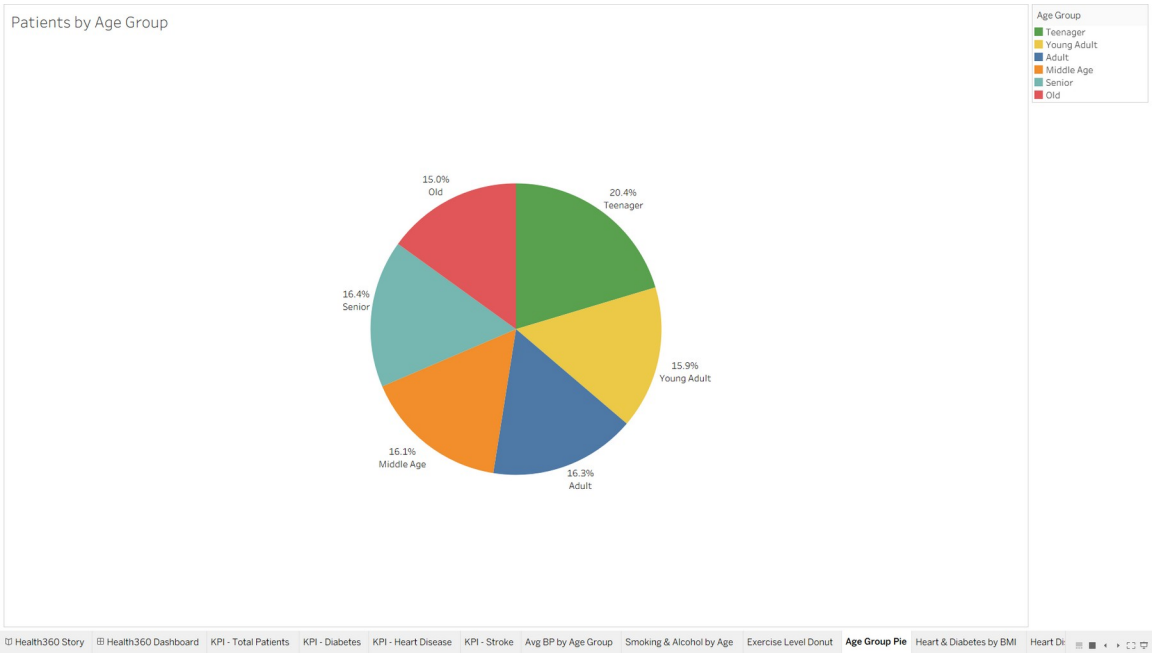


Figure 6. Heart Disease and Diabetes by BMI Category.

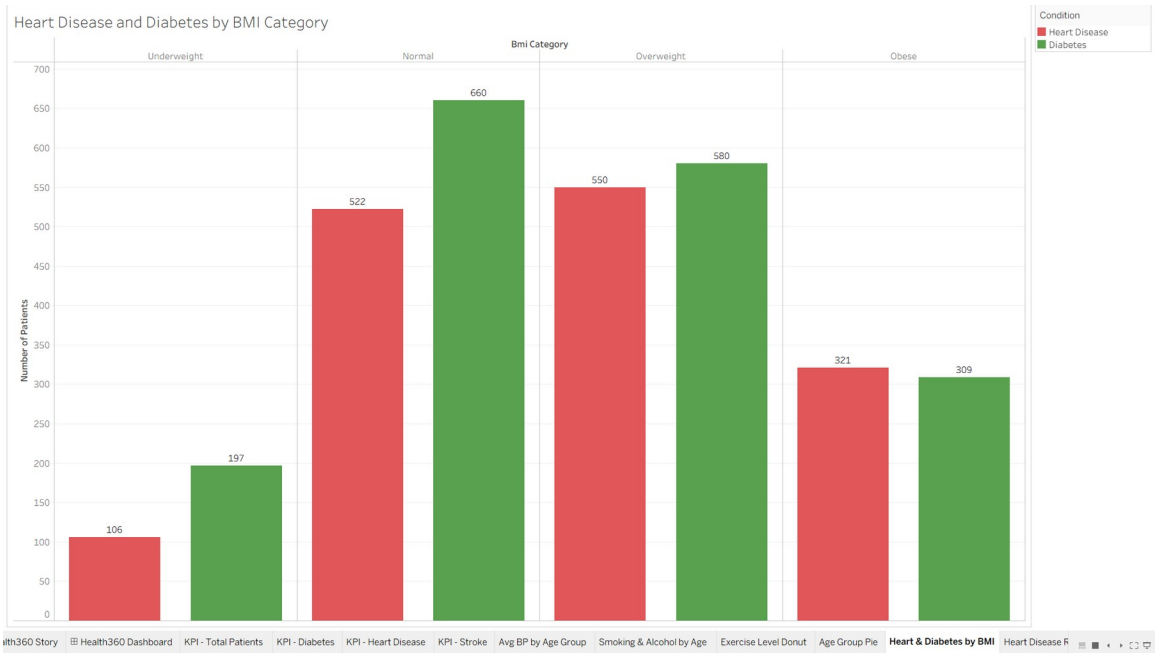


Figure 7. Heart Disease Rate by BMI Category: Underweight 20.5%, Normal 26.0%, Overweight 32.9%, Obese 40.0%.

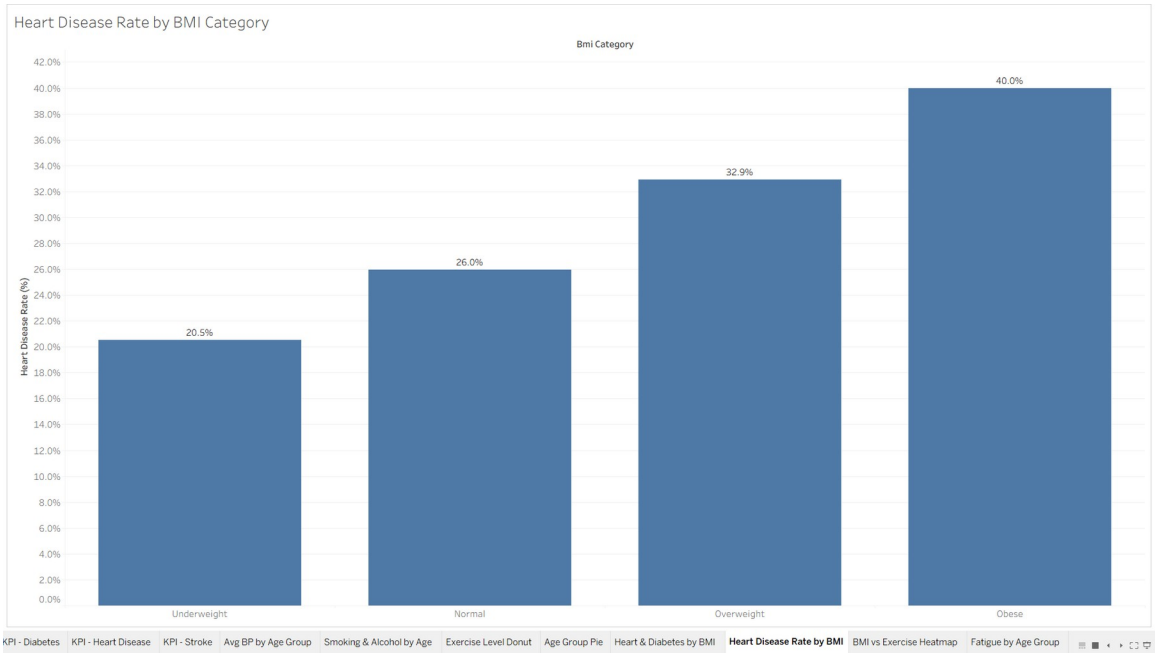


Figure 8. Heart Disease Rate by BMI and Exercise Level (heat map): highest 42.9% (Obese, Sedentary), lowest 18.8% (Underweight, Sedentary).

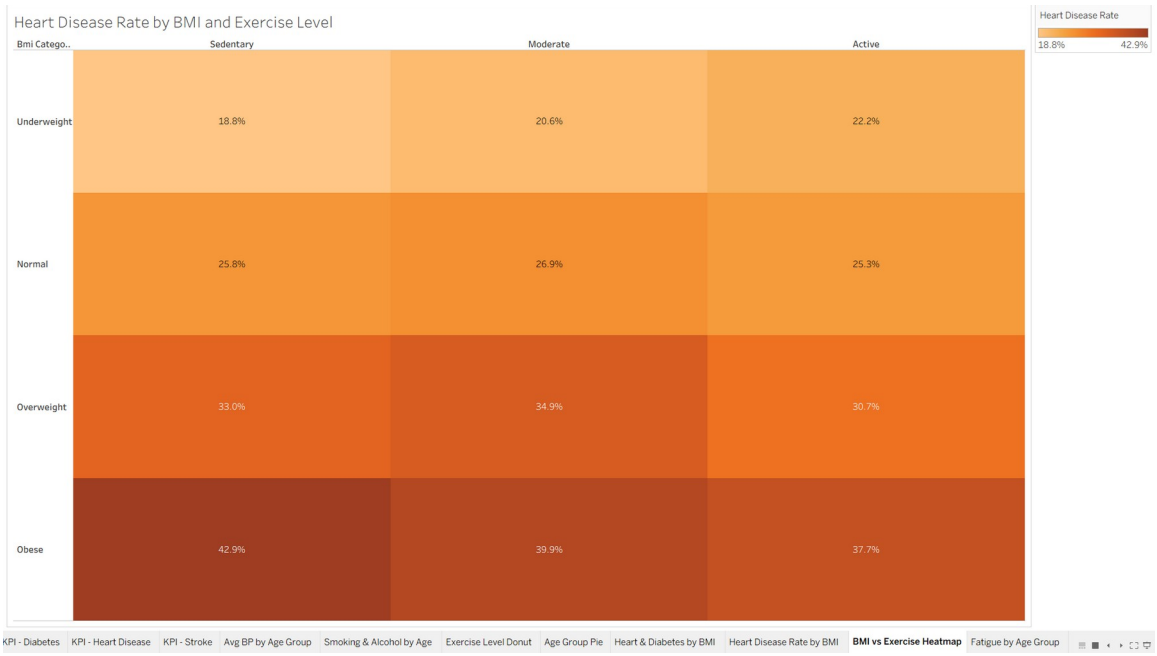


Figure 9. Patients by Age Group and Fatigue.

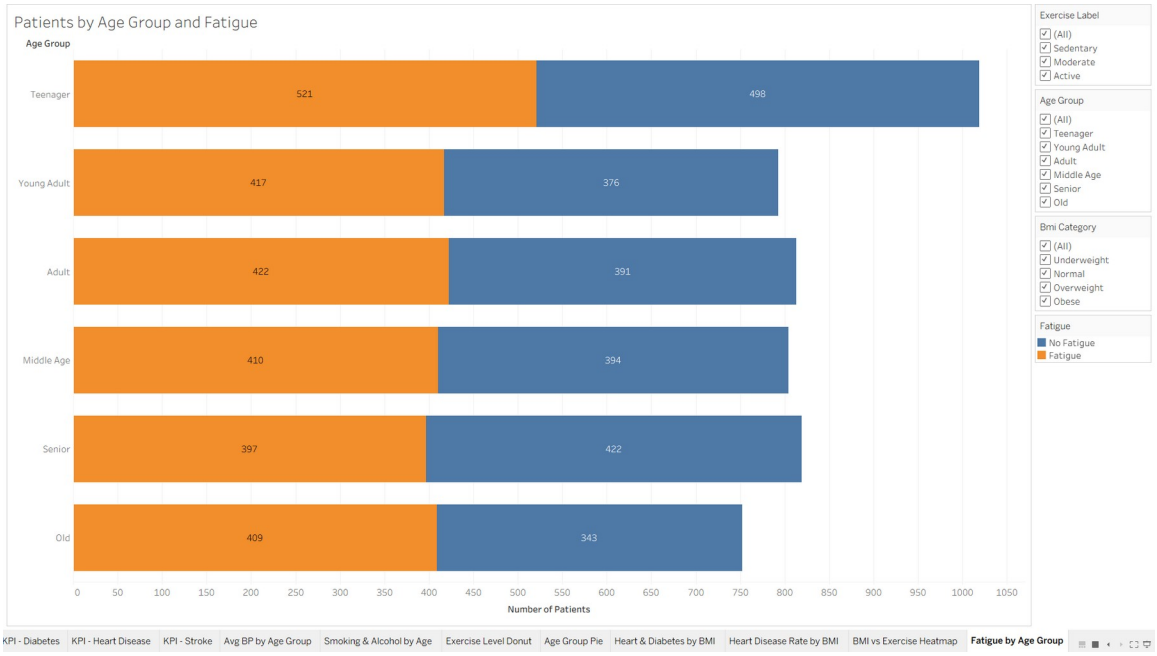


Figure 10. Health360 Story in Tableau (nine scenes).

Health360 Story

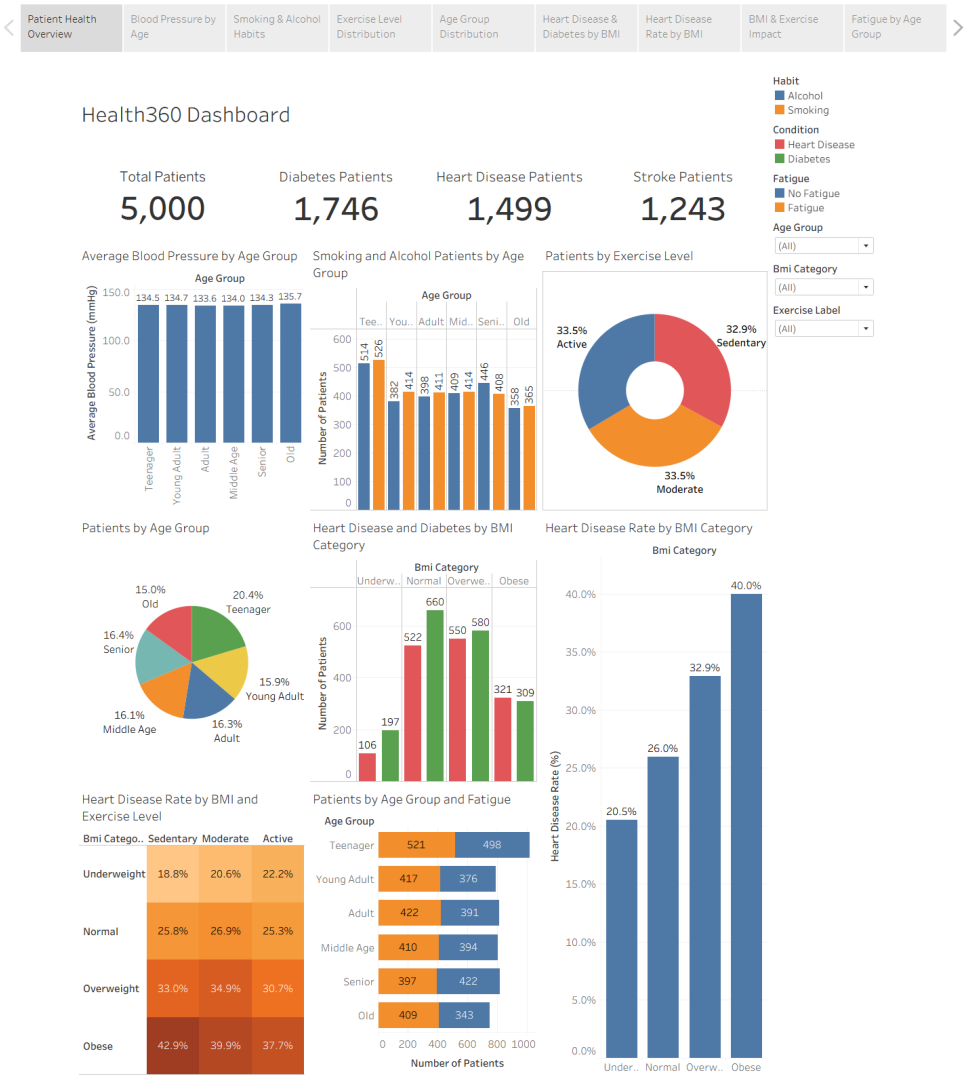


Figure 11. Flask web app: Home page.

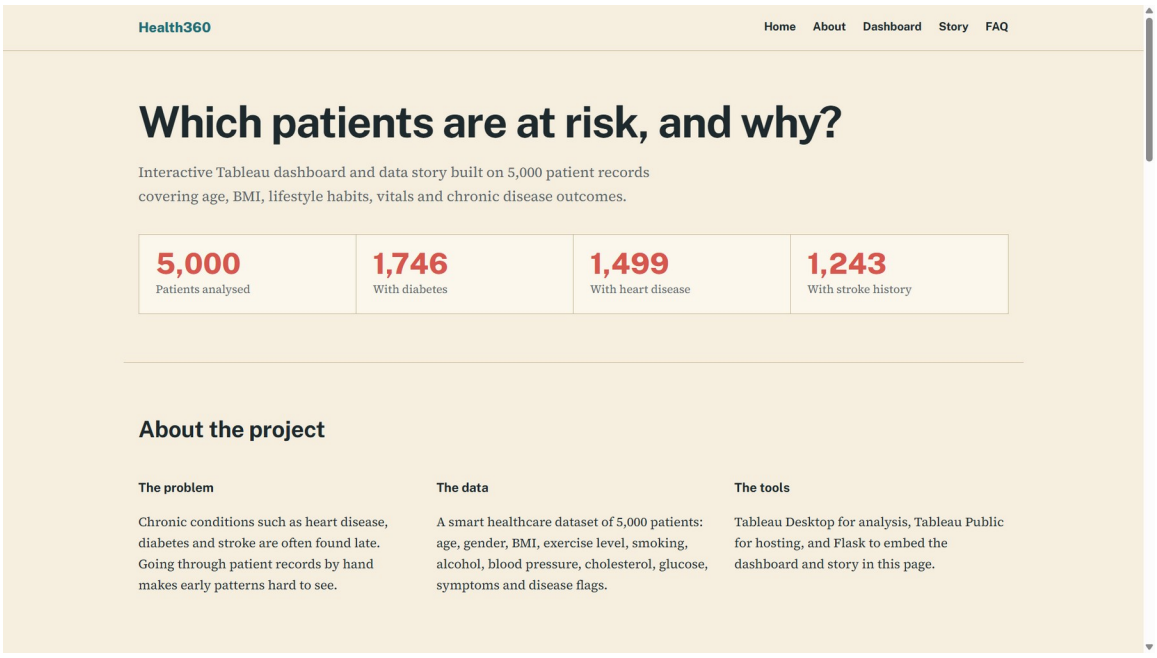


Figure 12. Flask web app: Dashboard section.

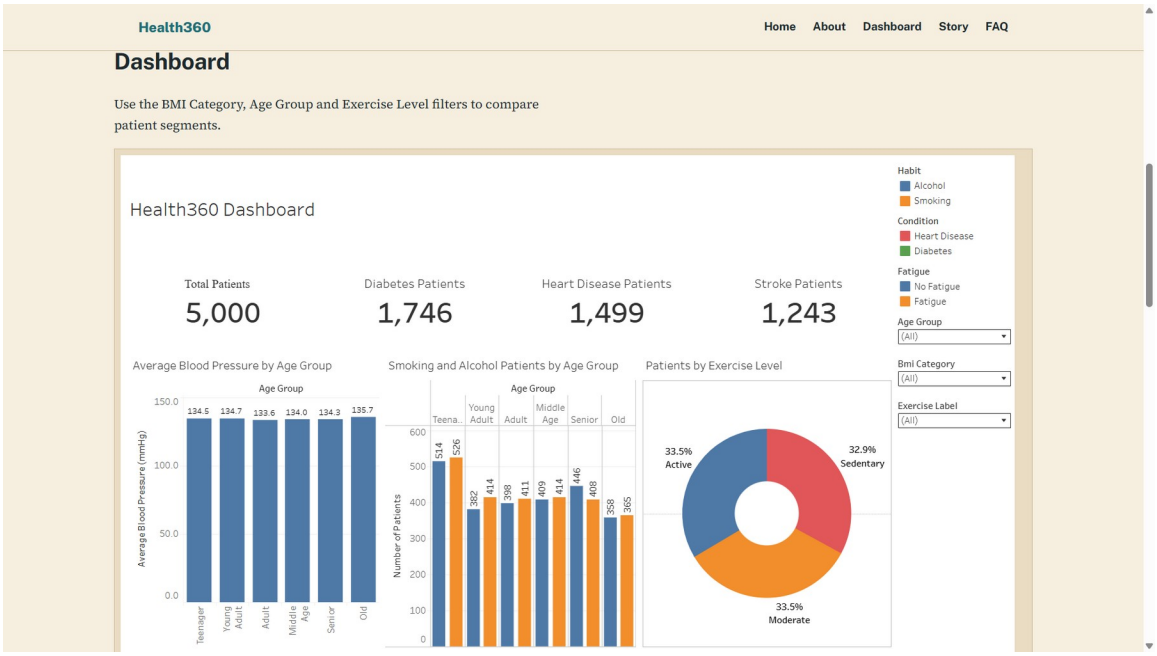
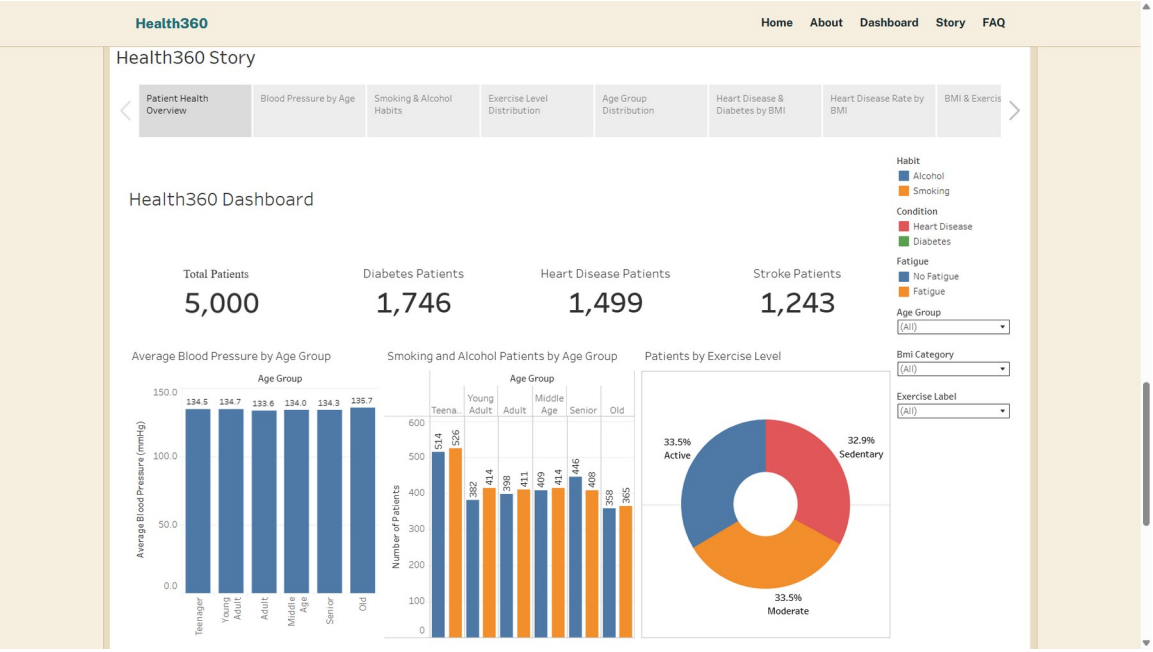


Figure 13. Flask web app: Story section.



Key findings

- Age is the dominant factor. No patient aged 50 or under has any of the three conditions. In the 51–60 group, 11.7% have heart disease and 21.8% have diabetes. In the 61–70 group, 79.7% have heart disease, 100% have diabetes and 60% have had a stroke. Every patient over 70 has all three.
- Heart disease rises steadily with BMI, from 20.5% (Underweight) to 40.0% (Obese).
- 54.9% of all patients have a blood pressure reading of 130 or above, so high blood pressure is widespread across age groups.
- Smoking, alcohol, cholesterol and glucose show very little relationship with the disease flags in this dataset.

The data looks synthetic, so read these patterns as a learning exercise and not as clinical evidence.

8. Advantages & Disadvantages

Advantages

- Filters let users compare patient segments in seconds.
- Calculated fields turn raw age and BMI into readable groups.
- Published on Tableau Public, so anyone with the link can view it.
- The Flask page makes the work easy to share and demo.

Disadvantages

- The dataset looks synthetic and has a constant health_risk_score column.
- Cross-sectional data shows association, not cause.
- Tableau Public workbooks are public, so real patient data must never be published this way.

9. Conclusion

Tableau turned raw patient records into clear, interactive views of disease risk. In this dataset age separates patients with and without chronic disease more than any other factor, BMI adds a steady secondary effect on heart disease, and high blood pressure is common at every age.

10. Future Scope

- Add a predictive model that scores each patient's risk.
- Use real, de-identified data with time tracking for each patient.
- Add more filters such as gender and symptoms.
- Host the Flask app online with a login for clinical staff.

11. Appendix

Source Code

The Flask app (app.py) serves one page and takes the two Tableau Public links as settings:

```
from flask import Flask, render_template

app = Flask(__name__)

DASHBOARD_URL = ""  # Tableau Public dashboard link
STORY_URL = ""      # Tableau Public story link

@app.route("/")
def home():
    return render_template("index.html",
                           dashboard_url=DASHBOARD_URL,
                           story_url=STORY_URL)

if __name__ == "__main__":
    app.run(debug=True)
```

Run it with `pip install flask` and `python app.py`, then open http://127.0.0.1:5000.

Dataset Link

https://drive.google.com/file/d/1RDbCMrWJaLQUufYcGbI6m9LgTvuJQcv2/view

GitHub & Project Demo Link

- Tableau Public dashboard: https://public.tableau.com/app/profile/nihar.vijayvargiya/viz/Health360_17934468105130/Health360Dashboard
- Tableau Public story: https://public.tableau.com/app/profile/nihar.vijayvargiya/viz/Health360_17934468105130/Health360Story

- GitHub repository: https://github.com/NiharVijayvargiya/Health360-Healthcare-Dashboard
- Demo video: https://drive.google.com/file/d/1HwZMmzwnsz3MI2uoAWj93sWubI4FbPbZ/view?usp=drive_link